

Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in $M \geq 6.5$ Earthquake Catalogs, 1973–2026 CE

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Abstract.

We analyse an analysis catalog of 4,267 unique $M \geq 6.5$ events (modern window 1973–2026: 2,041 events; provenance: 4,418 CSV rows, ~151 NOAA $M < 6.5$ excluded from clustering) using the Baiesi-Paczuski metric η with tectonic-path distance (Bird 2003). The detector yields 47 algorithmic candidates (27 modern); historical NOAA records ($n=47$) reported in Appendix A only. Primary ETAS null (literature H&S 2003, decoupled): $\text{mean} \approx 15.4$, $p_{\text{ETAS}} \leq 0.001$ — exceeds local aftershock expectation; not teleseismic proof (Sec. 5.4). Secondary diagnostic (catalog WLS): $p_{\text{ETAS}} = 1.0$, $\text{mean} = 27.0 = N_{\text{obs}}$ — detector--calibration coupling. GK mainshocks: $N=27$ unchanged. Global-series hypothesis not supported (Sec. 5.4–5.6). Permutation $p=0.0001$ (1/10,001) rejects temporal Poisson null only. Limitations — Sec. 5.6.

Keywords: global seismicity; seismic series; earthquake clustering; heuristic metric with tectonic hint; Baiesi-Paczuski metric; ETAS validation; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra-Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that $M \geq 7$ clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global $M \geq 7$ and $M \geq 8$ rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi-Paczuski [2004] metric and Zaliapin-Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. Test (and if warranted, falsify) the hypothesis that physically meaningful multi-regional global series exist, with primary inference on the modern window (1973–2026), using complementary null tests (permutation vs ETAS) and explicit detector liberalness assessment. Scope. We combine nearest-neighbor clustering with heuristic tectonic hint and ETAS validation; this complements global rate tests (Michael 2011; Shearer & Stark 2012) with a different eta-linkage statistic but does not supersede their conclusions.

2. DATA AND METHODS

2.1 Catalog compilation

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records pre-1900, 47 events). Duplicates were merged using ± 30 days and ≤ 50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains 4,267 unique $M \geq 6.5$ events (from 4,418 CSV rows; ~ 151 $M < 6.5$ excluded from clustering). USGS ComCat: 2,088 raw (1973–2026) $\rightarrow$ 2,041 after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). `quality_score` is metadata, not an inclusion filter. Primary analysis set: events 1900–2026 (4,218); 47 pre-1900 records remain in CSV (provenance) but are excluded from the primary detector pipeline and ETAS calibration window (1973–2026 only) — see Appendix A. Final catalog: 4,267 $M \geq 6.5$ events (4,418 CSV rows).

2.1.1 Catalog completeness

Maximum-curvature analysis yields $M_c = 6.55$. Maximum-likelihood b-value from 1,688 events above M_c :

$$b = 0.911 \pm 0.018 \quad (n = 1,688 \text{ events})$$

2.2 Heuristic metric with tectonic hint

We tested an alternative to Euclidean distance (Bird 2003 graph). In 98% of pairs the implementation falls back to 1.5x great-circle distance — effectively scaled Euclidean; no improvement for global analysis (failed hypothesis test). r_{ij} is the shortest path between hypocenters along the Bird (2003) plate-boundary graph (20 key segments). Paths are computed with Dijkstra's algorithm (NetworkX). If either hypocenter lies >500 km from the nearest boundary node, or no graph path exists, $r_{ij} = 1.5 \times r_{GC}$ (great-circle Haversine). Fallback audit (analyze_tectonic_fallback.py, fig07): 4987 pairs from 500 events; 98.0% GC fallback, 2.0% Dijkstra (4015 snap>500 km, 872 no path, 100 Dijkstra; 95 materially different; examples FE31 Japan, FE35 Philippines). Metric adds value for ~2% boundary-proximal pairs only. Tectonic diagnostic: median $\Delta \log_{10} \eta = +0.28$.

$$\eta_{ij} = t_{ij} * r_{ij}^{1.6} * 10^{(-b * m_i)}$$

$b=1.0$ — deliberate Baiesi & Paczuski (2004) simplification; catalog $b=0.911 \pm 0.018$ for M_c /completeness and MC null only — not in the η formula. Zaliapin (2008): $D \approx b - \eta$, η_{a_0} , cluster shifts not tested.

2.3 Analysis pipeline

Raw catalogs → dedup → exclude $M < 6.5$ (~151 NOAA) → GK declustering (primary) → η NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — different algorithms (GK: magnitude-dependent windows; ZBZ: η NN + KDE threshold). At global $M \geq 6.5$ scale ZBZ is permissive (sparse events → high η). GK is sole primary for inference; ZBZ sensitivity only, does not replace GK. If ZBZ were primary, $N=27$ likely unchanged (untested).

Why GK-ZBZ counts differ: not contradictory methods — GK removes short-range fore/aftershocks via fixed windows; ZBZ classifies only 1 event with exceptionally low η at global catalog scale. Minimal ZBZ removal confirms declustering is not a critical fork.

2.4 Clustering and detector criteria

1. GK declustering (primary) → mainshocks for η NN forest.
2. η NN forest: $b=1.0$, $r^{1.6}$; tectonic Bird 2003 (1.5x GC fallback).
3. Sliding windows 1, 2, 5 yr (1-yr step); merge 142→47.
4. Detector gate: $N \geq 4$; $M \geq 6.5$; mean pairwise GC > 1500 km.
5. Flinn-Engdahl zone count — diagnostic only (not admission criterion).

2.5 Threshold η_0 and ETAS calibration

η_0 : KDE valley in $\log_{10}(\eta)$ (Zaliapin & Ben-Zion 2013); see figures/grl/fig_eta_threshold.png (scripts/plot_eta_threshold.py). KDE stability at global $M \geq 6.5$ not verified.

ETAS calibration (scripts/calibrate_etas.py, results/etas_calibration.json): μ = GK mainshocks/T (closed form); Omori c, p via scipy.minimize Nelder-Mead ($c=10^{-4}$ at lower bound); K, α via WLS (numpy.linalg.lstsq) on 24 aftershocks (≤ 500 km). All c in days (ETAS standard); lower bound 10^{-4} day (≈ 8.6 s) is a numerical floor; literature c typically 0.001–0.01 day. $K \approx 0.495$ vs literature ~ 0.08 — simplified WLS, not Ogata MLE. Parameter CIs not estimated. H&S 2003 comparison: $\mu=0.008$, $K=0.08$, $\alpha=1.0$, $c=0.005$, $p=1.1$.

2.6 Statistical validation

Permutation test: $n = 10,000$, $p \leq 0.0001$, $z = -6.17$ (modern). ETAS validation: calibrated params ($\mu \approx 0.103$, $K \approx 0.495$, $\alpha \approx 0.063$, $c \approx 10^{-4}$ d, $p \approx 1.36$); 1000 catalogs/seed. Single seed=42: FPR=1000/1000; mean 27.0, $p_{\text{ETAS}} = 1.0$; $N_{\text{obs}}=27$. Multiseed ETAS (seeds 42–51, $n=1000$ each, results/etas_multiseed.json): mean=27.0, $\sigma=0.0$, FPR=1.0, $p_{\text{ETAS}}=1.0$ for all 10 seeds — perfect stability because calibrated ETAS matches catalog event rate (~ 2001 background events). Literature H&S 2003: mean ≈ 15.4 , $p_{\text{ETAS}} \leq 0.001$ — $N_{\text{obs}}=27$ exceeds null. Dual ETAS null: lit. ($\mu=0.008$, $K=0.08$) $\rightarrow$ mean ≈ 15.4 , $p \leq 0.001$; calibrated ($\mu \approx 0.103$, $K \approx 0.495$) $\rightarrow$ mean=27.0, $p=1.0$ (detector coupling). $\alpha=b=0.911$ (results/etas_validation_b0911.json): $p_{\text{ETAS}}=1.0$, mean ≈ 27 . Detector liberalism — see Sec.5.6 (FPR=1000/1000). Multiple comparisons (Methods): BH post-hoc on $N=47$, 45/47 at $q=0.05$ — not discovery. Declustering: GK 2,017/2,041 (primary); ZBZ 2,040/2,041 (sensitivity only).

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 detector candidates (not validated global series): 27 modern ($p \leq 0.0001$), 15 early instrumental ($p = 0.007$; pre-1960 incompleteness caveat), 5 historical candidates (not significant, $p = 0.46$; 47 $M \geq 6.5$ events pre-1900). 142 cluster candidates before filtering.

ID	N	Reg.	Mmax	Period	lat	lon	Notes
1905-1910	193	43	8.8	1905–1910	-60..72	-180..180	Early instrumental; incomplete before ~ 1960
S047	53	5	8.0	1982–2024	-21.5..18.9	-175.5..155.2	Western Pacific
S170	46	12	9.1	2002–2023	-14.3..33.8	-113.1..167.3	Sumatra 2004 (M 9.1)

S095	25	4	7.9	1989–2017	-8.1..16.5	120.4..14 6.4	Western Pacific arc
S116	22	5	8.2	1993–2021	-31.7..10. 8	-179.5..1 79.4	South Pacific

Table 1. Top-5 detector candidates (not ETAS-validated physical series).

3.2 Spatial-temporal distribution

Elevated detector candidate frequency (not validated physical series) occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic diagnostic: median $\Delta\log_{10}\eta = +0.28$ on random pairs (mostly 1.5x GC fallback).

4. DISCUSSION

The detector is liberal: catalog-matched ETAS $\text{mean}=27.0=N_{\text{obs}}$ ($p_{\text{ETAS}}=1.0$) indicates detector--calibration coupling; literature ETAS $\text{mean}\approx 15.4$, $p_{\text{ETAS}}\leq 0.001$ — N_{obs} exceeds local-clustering null. No physical mechanism; tectonic metric failed (98% GC fallback). Compatible with Michael (2011) and Shearer & Stark (2012).

5. CONCLUSIONS

The detector is liberal: matched ETAS calibration yields $p_{\text{ETAS}}=1.0$ (coupling); literature ETAS yields $\text{mean}\approx 15.4$, $p_{\text{ETAS}}\leq 0.001$. Global-series hypothesis not supported (Sec. 5.4–5.5); ETAS null is parameter-sensitive. Permutation rejects Poisson times only (Ogata 1988).

1. Heuristic with tectonic hint: 98% GC fallback — failed hypothesis test.
2. 47 detector candidates indistinguishable from ETAS null; Delta CFS/dynamic stress — future work only.

Future work / supplement: Static Delta CFS and dynamic stress for S170, S047, S095; full ETAS MLE and multi-seed robustness.

APPENDIX A. PRE-1900 NOAA RECORDS

Forty-seven fragmentary paleoseismic/historical $M\geq 6.5$ records from NOAA NGDC are retained in data/processed/unified_catalog_full.csv for provenance (not removed). These 47 events are excluded from the primary detector pipeline and ETAS calibration window: pipeline_v2.py and calibrate_etas.py use the modern catalog 1973–2026 only ($N=2,041$). Epoch-stratified counts include pre-1900 descriptively via run_full_historical_analysis.py but do not enter primary significance claims.

DATA AND CODE AVAILABILITY

Data and code: github.com/marshalkin-ux/paleoseismic-clustering (docs/data_availability.md). External DOI deposition (Zenodo) deferred — GitHub only.

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